

PORT MACQUARIE, Australia

WMO: 947860

Lat: 31.4336S

Lon: 152.8656E

Elev: 16

StdP: 14.69

Time Zone: 10.00 (AUS)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.7	40.4	29.7	23.8	54.9	32.2	26.7	52.3	22.3	57.9	20.0	59.3	2.4	300	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.0	86.9	73.2	84.3	72.5	82.3	72.0	76.0	83.3	74.7	81.2	73.6	79.7	12.3	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.8	125.8	79.6	72.7	121.4	78.6	71.6	116.8	77.3	39.5	83.7	38.3	81.4	37.2	79.8	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.0	18.5	16.5	DB	32.2	97.4	2.0	6.2	30.7	101.9	29.5	105.5	28.4	109.0	26.9	113.6
			WB	31.4	78.3	1.8	1.4	30.2	79.3	29.1	80.2	28.1	80.9	26.9	82.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	73.3	72.9	70.7	65.8	60.4	56.7	54.6	55.8	60.1	63.8	67.5	70.8	(6)
(7)		DBStd	7.53	3.73	3.46	3.10	3.10	3.73	3.56	3.32	3.73	4.27	4.47	4.53	4.10	(7)
(8)		HDD50	9	0	0	0	0	0	2	4	3	0	0	0	0	(8)
(9)		HDD65	1294	0	0	1	25	147	249	323	288	158	77	23	3	(9)
(10)		CDD50	5235	722	640	641	475	324	203	146	182	303	429	525	645	(10)
(11)		CDD65	1043	257	220	177	51	5	0	0	1	11	40	98	183	(11)
(12)		CDH74	6042	1682	1306	807	204	22	1	3	19	129	279	534	1056	(12)
(13)		CDH80	1116	356	253	101	13	0	0	0	3	28	51	105	206	(13)
(14)	Wind	WSAvg	7.4	8.6	8.3	7.4	6.6	6.4	6.2	6.8	7.3	8.0	8.5	8.7	(14)	
(15)	Precipitation	PrecAvg	59.9	6.0	6.3	7.6	6.7	5.7	5.6	2.7	2.4	2.3	4.0	4.7	5.1	(15)
(16)		PrecMax	101.7	21.7	13.9	26.7	24.4	15.0	18.6	11.3	10.2	7.8	15.6	11.1	14.7	(16)
(17)		PrecMin	31.4	1.1	1.1	0.4	0.3	0.3	0.3	0.0	0.0	0.0	0.5	0.5	0.7	(17)
(18)		PrecStd	17.0	4.6	3.2	5.3	5.7	3.9	3.9	2.3	2.3	1.7	3.2	2.6	3.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	90.8	90.5	86.2	81.7	76.5	72.0	73.2	76.5	84.5	85.4	87.9	90.1	(19)
(20)			MCWB	75.5	76.2	73.9	68.6	64.1	60.9	58.4	58.8	65.2	66.6	70.7	73.0	(20)
(21)		2%	DB	86.1	85.2	82.5	78.4	73.6	69.2	68.7	71.5	77.4	80.0	82.7	84.6	(21)
(22)			MCWB	74.7	74.5	72.7	67.5	63.5	59.7	57.4	57.5	62.9	66.4	70.5	72.3	(22)
(23)		5%	DB	83.5	82.8	80.4	76.2	71.4	67.1	66.0	68.8	73.8	76.8	79.3	81.9	(23)
(24)			MCWB	73.3	73.0	71.3	66.5	61.8	58.3	56.1	57.0	61.9	65.7	69.2	71.5	(24)
(25)		10%	DB	81.5	80.8	78.5	74.3	69.2	65.1	63.8	66.4	70.9	74.0	76.6	79.5	(25)
(26)			MCWB	72.2	71.7	69.9	65.4	60.6	57.3	54.9	55.7	60.5	64.5	67.9	70.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.4	77.8	76.0	71.6	68.0	64.3	63.0	63.5	67.9	71.1	74.4	76.1	(27)
(28)			MCDWB	86.3	86.0	83.0	77.5	72.8	69.2	67.6	70.7	77.1	79.0	83.4	84.1	(28)
(29)		2%	WB	76.0	76.1	74.0	69.6	66.0	62.1	60.5	60.9	65.7	68.9	72.1	74.2	(29)
(30)			MCDWB	83.4	82.8	79.8	75.7	70.7	66.4	64.6	67.1	73.3	76.2	79.4	81.5	(30)
(31)		5%	WB	74.7	74.5	72.7	68.2	64.1	60.6	58.5	59.2	64.2	67.3	70.5	72.8	(31)
(32)			MCDWB	81.1	80.3	78.0	73.9	68.5	64.8	63.3	65.5	71.1	74.3	77.0	79.4	(32)
(33)		10%	WB	73.6	73.1	71.5	66.8	62.4	59.0	56.7	57.6	62.6	65.8	69.1	71.5	(33)
(34)			MCDWB	79.6	78.7	76.6	72.3	67.0	63.2	61.9	64.0	68.8	72.2	74.9	77.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.0	14.6	15.0	16.9	18.3	17.7	20.0	22.2	22.1	19.1	16.2	15.8	(35)
(36)			MCDBR	18.2	18.1	18.1	20.2	21.3	20.5	24.1	27.1	26.9	24.1	20.9	19.5	(36)
(37)			MCWBR	8.4	8.6	9.4	10.7	11.6	12.2	14.1	15.0	14.5	12.5	10.3	9.2	(37)
(38)		5% WB	MCDBR	15.7	16.3	15.9	18.0	16.9	16.3	18.7	22.1	22.9	20.1	17.4	17.0	(38)
(39)			MCWBR	7.9	8.2	8.8	10.4	10.4	10.8	12.3	13.6	13.6	11.4	9.2	8.6	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.396	0.379	0.359	0.341	0.317	0.309	0.301	0.312	0.352	0.367	0.386	0.395	(40)	
(41)		taud	2.472	2.542	2.593	2.607	2.631	2.642	2.651	2.603	2.486	2.480	2.472	2.466	(41)	
(42)		Ebn at Noon	299	298	294	283	276	271	279	289	292	299	300	301	(42)	
(43)		Edh at Noon	37	34	30	27	23	22	23	26	33	36	37	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2089	1850	1562	1291	1030	845	968	1268	1642	1845	1966	2074	(44)	
(45)		RadStd	149	144	130	86	73	62	78	82	102	154	189	167	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			+0.94	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (10 neighbors)			+0.72	N/A	N/A	N/A	N/A	N/A	-108	+288	+155

Nomenclature: See separate page